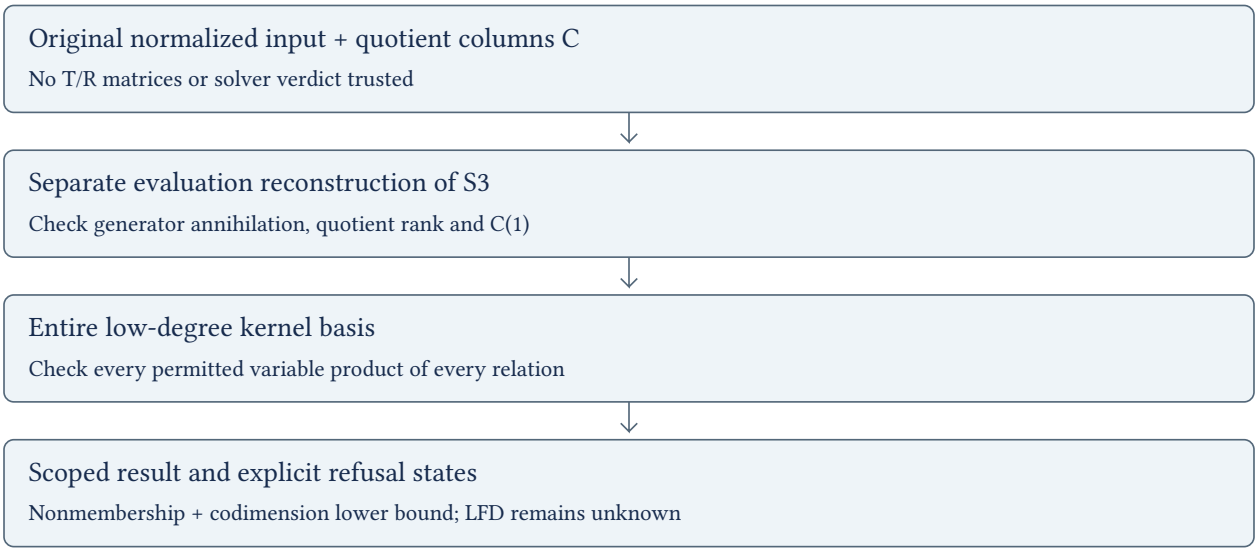
Direct closed-kernel checking and SAT extension bounds

2026-11-09. Written proof review and fixed software controls.

Checked implementation: The bounded CLI directly reconstructs the quadratic coordinate equations and tests a basis of the entire eligible kernel. A map satisfying the input equations but failing kernel multiplication is rejected. Eight software controls pass. Acceptance, predicate rejection and capacity refusal have distinct retained JSON artifacts. No search solver ran.



Derived SAT bound: For one quadratic definitional extension G of F with M greater than zero solutions, let b = max(2, floor(log2(M+1))). The reviewed arguments give dF <= max(dG,b)+2 and dG <= max(dF,b+1). Consequently uniformly bounded solution counts permit transfer of LFD boundedness or unboundedness.

Scope of the counterexample: A generic family with a degree-n input monomial has LFD zero before extension and at least n after adding one quadratic definition. That family rules out unrestricted additive invariance. Its high-degree generator means it does not settle the quadratic Semaev case.

Remaining obligation: Construct growing full-closure witnesses for actual Semaev inputs and supply the required UNSAT, solution-count or missing-consequence bridge. No infinite hard Semaev family has been established.

Evidence: direct-checker-audit.md; direct-checker-final-validation/receipt.json; sat-one-bit-elimination.md; sat-one-bit-elimination-review.md. Input limits and vector-operation accounting are explicit; they are not native instruction measurements. The first test-file path failure and earlier source snapshots are preserved.